

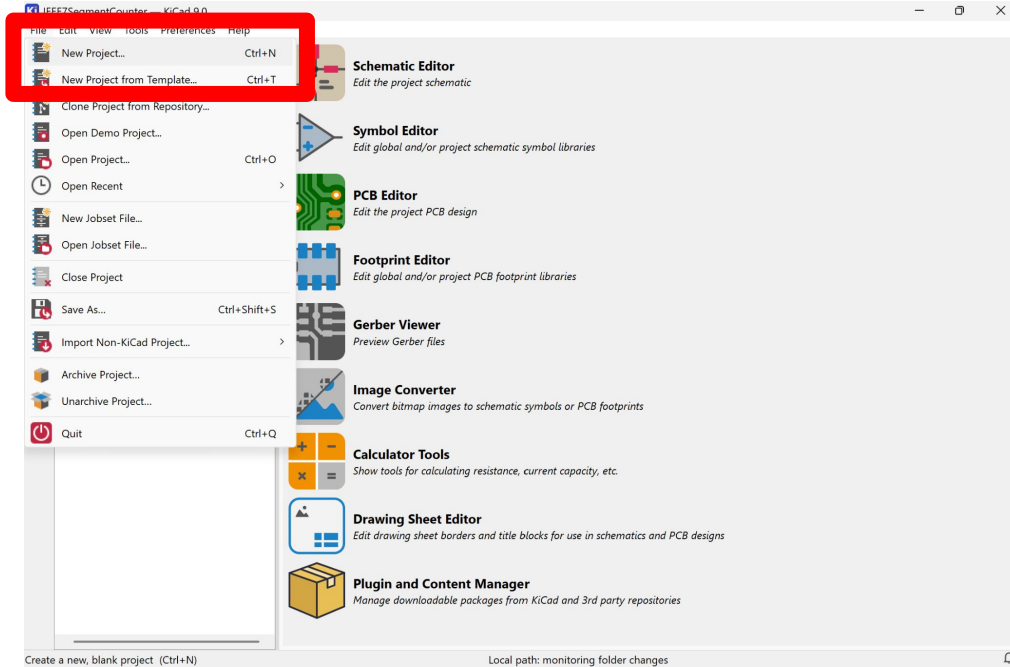
KiCad Workshop

Go to <https://github.com/psuieeee> for completed files

Attendance

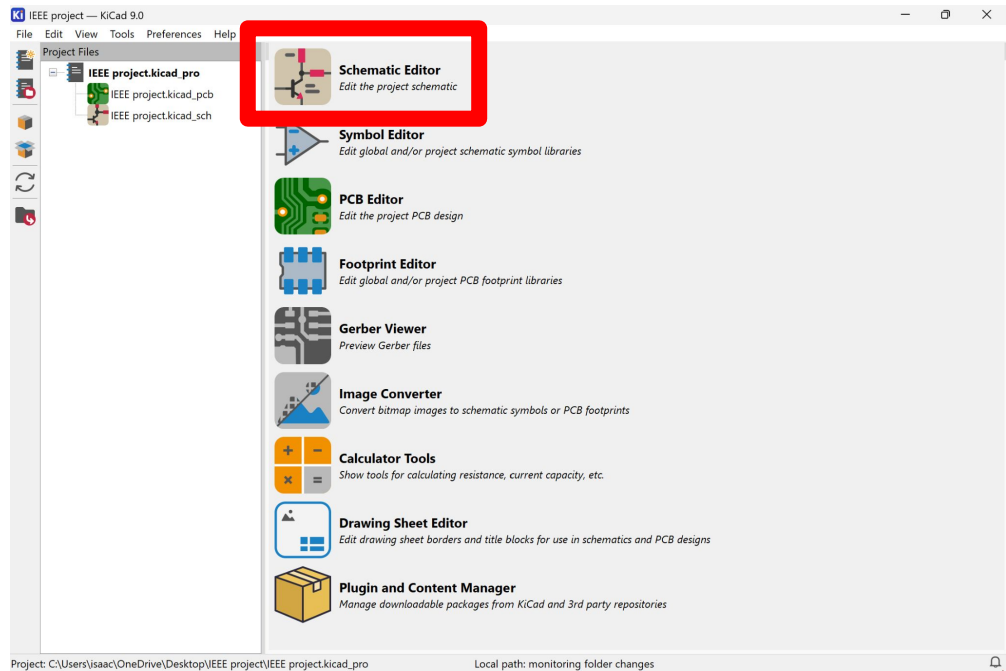


Create a project

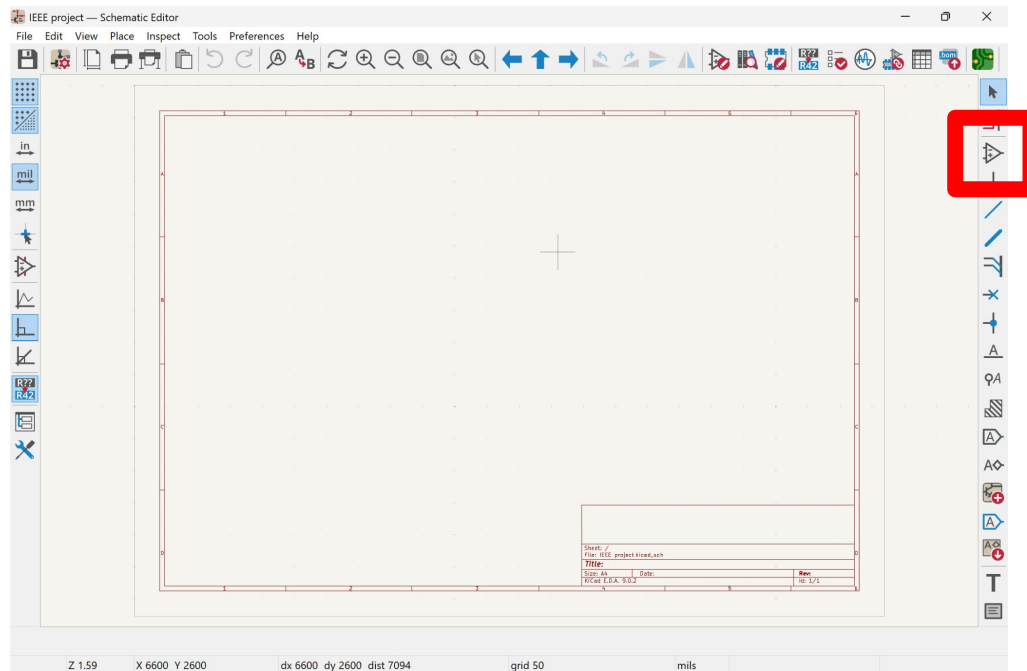


Open Schematic Editor

Open Schematic Editor



Place Symbols



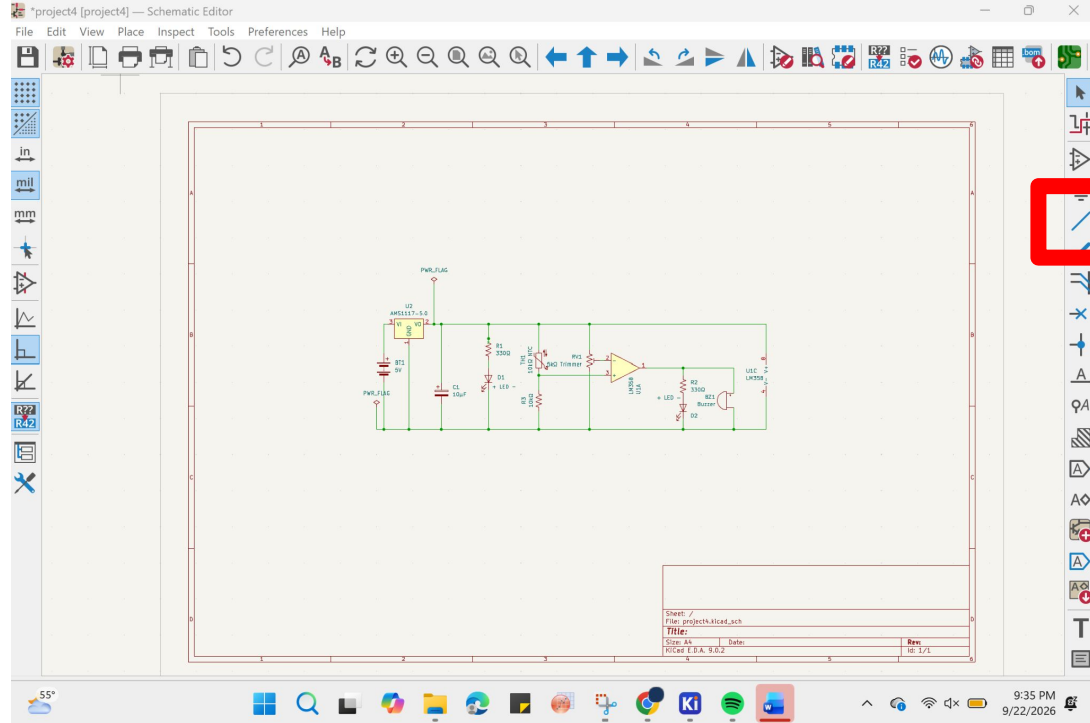
Shortcut (A)

Required Symbols

Symbol Name	Details
R_US	330 (x2), 10k ohm
C_Polarized	10 uF
Thermistor_PTC	5k ohm
Battery	5V
AMS1117-5.0	
LED	x2
R_Potentiometer_Trim_US	5k ohm
LM358 (Unit A & C)	
Buzzer	
PWR_Flag	x2

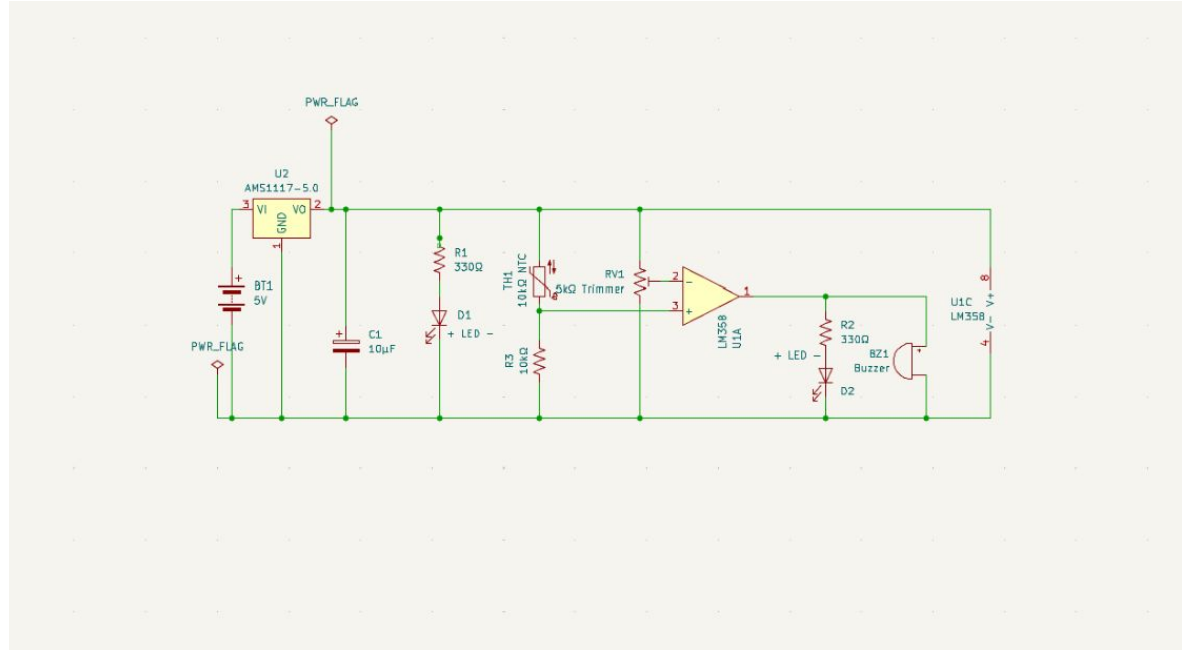
Shortcut (A)

Align Components & Wiring

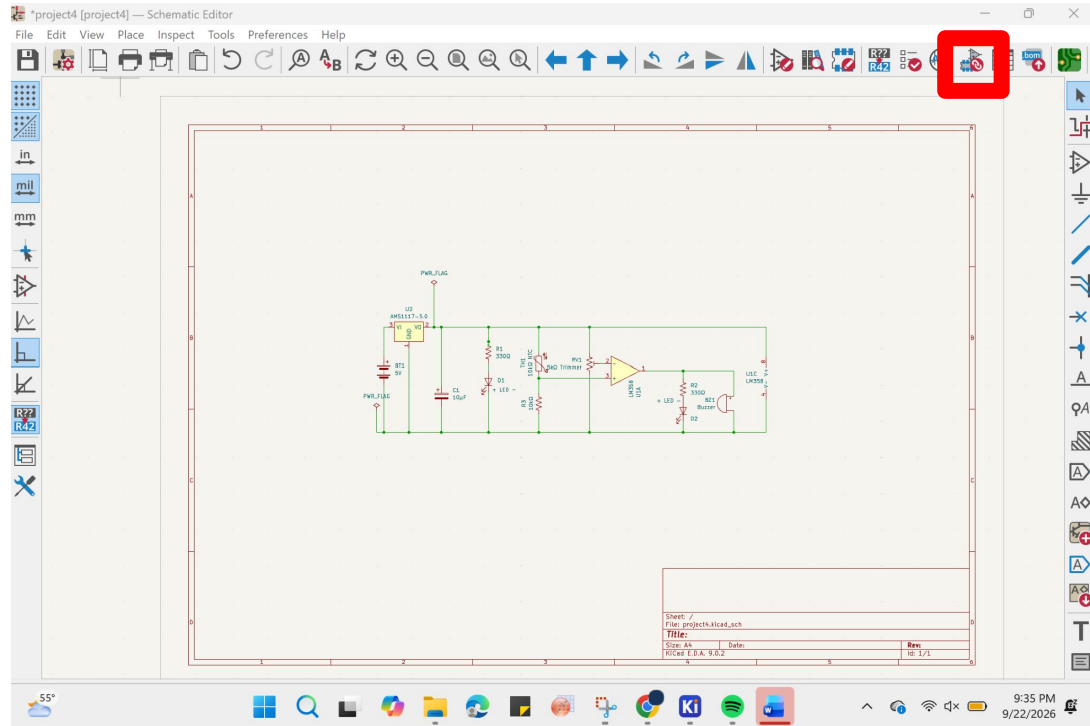


Shortcut (W)

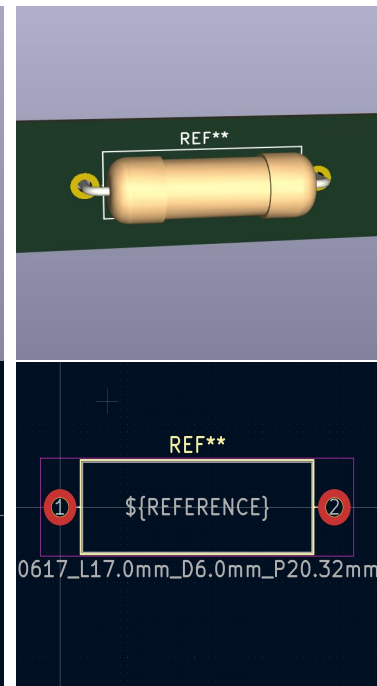
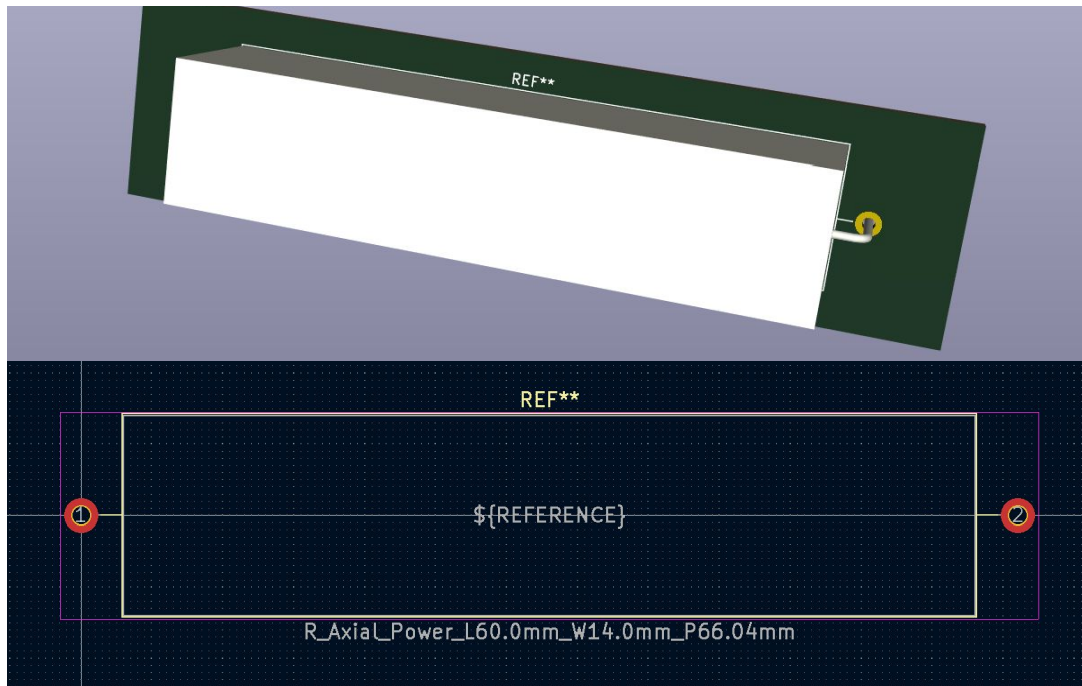
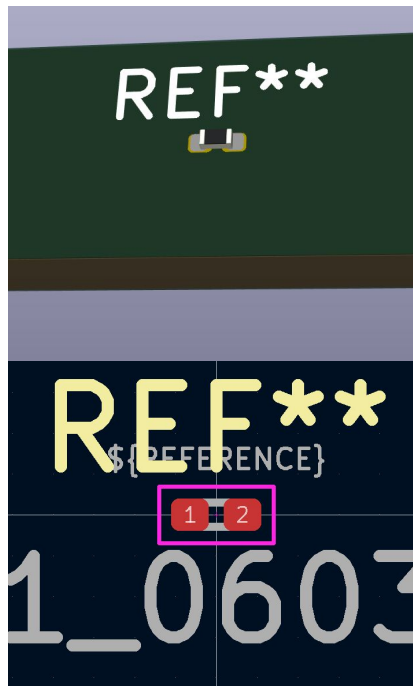
Circuit Diagram



Adding Footprints



Same Device, Different Footprint

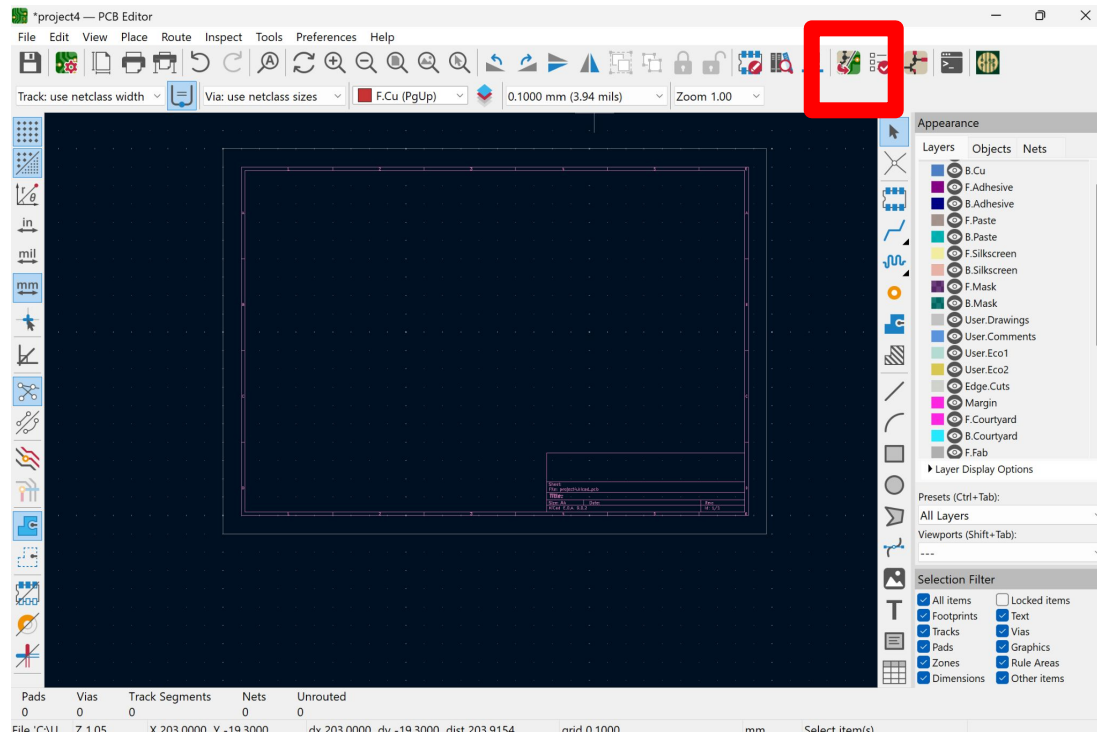


Adding Footprints

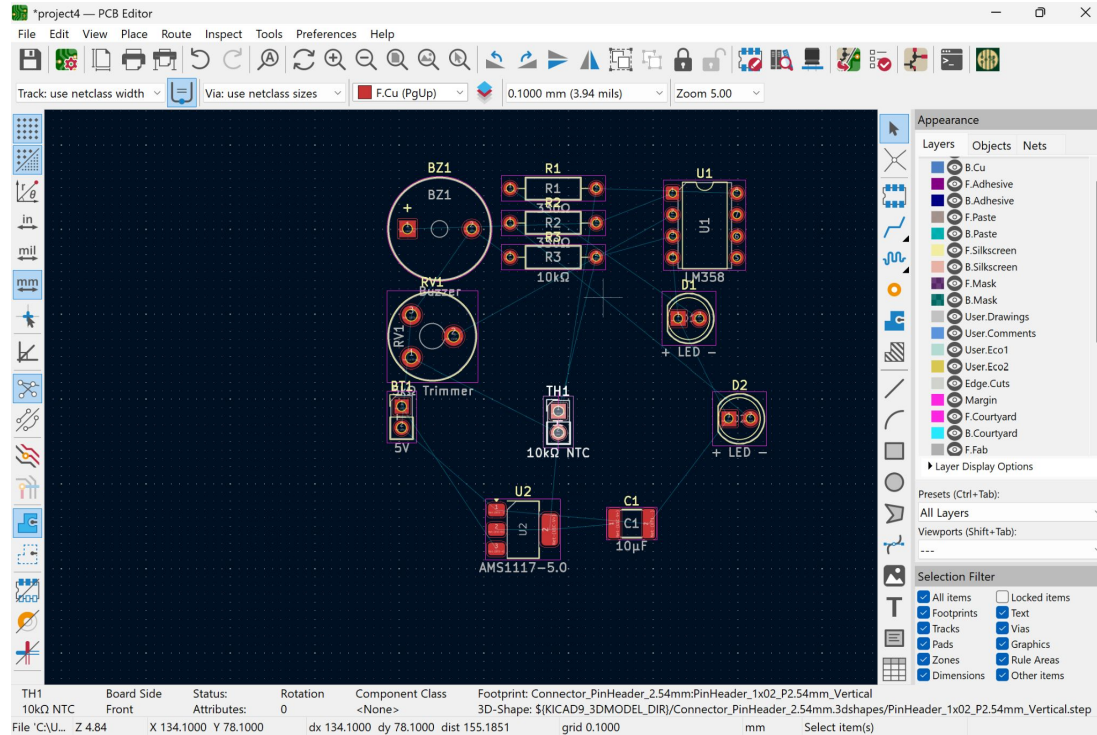
Symbol Name	Footprint
BT1	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
BZ1	Buzzer_Beeper:Buzzer_12x9.5RM7.6
C1	Capacitor_SMD:C_1812_4532Metric
D1	LED_THT:LED_D5.0mm
D2	LED_THT:LED_D5.0mm
R1	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal

Symbol Name	Footprint
R2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
R3	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
RV1	Potentiometer_THT:Potentiometer_Piher_PT-10-V05_Vertical
TH1	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
U1	Package_DIP:DIP-8_W7.62mm
U2	Package_TO_SOT_SMD:SOT-223-3_TabPin2

Placing Footprints

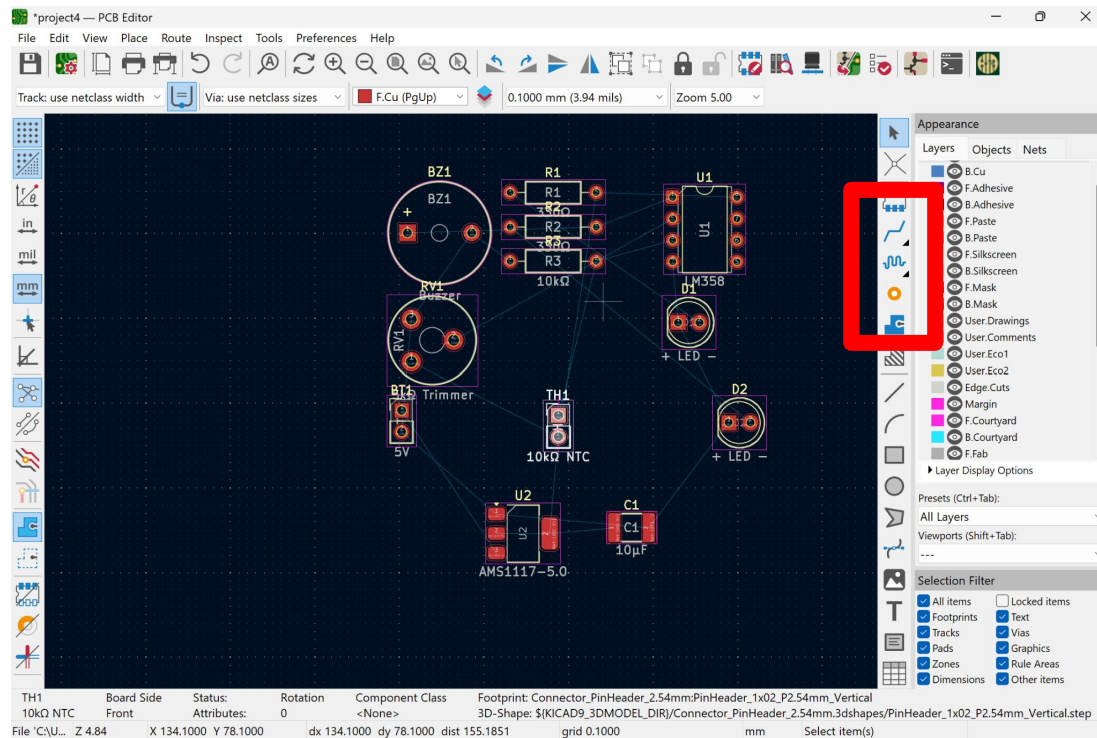


Arrange Footprints



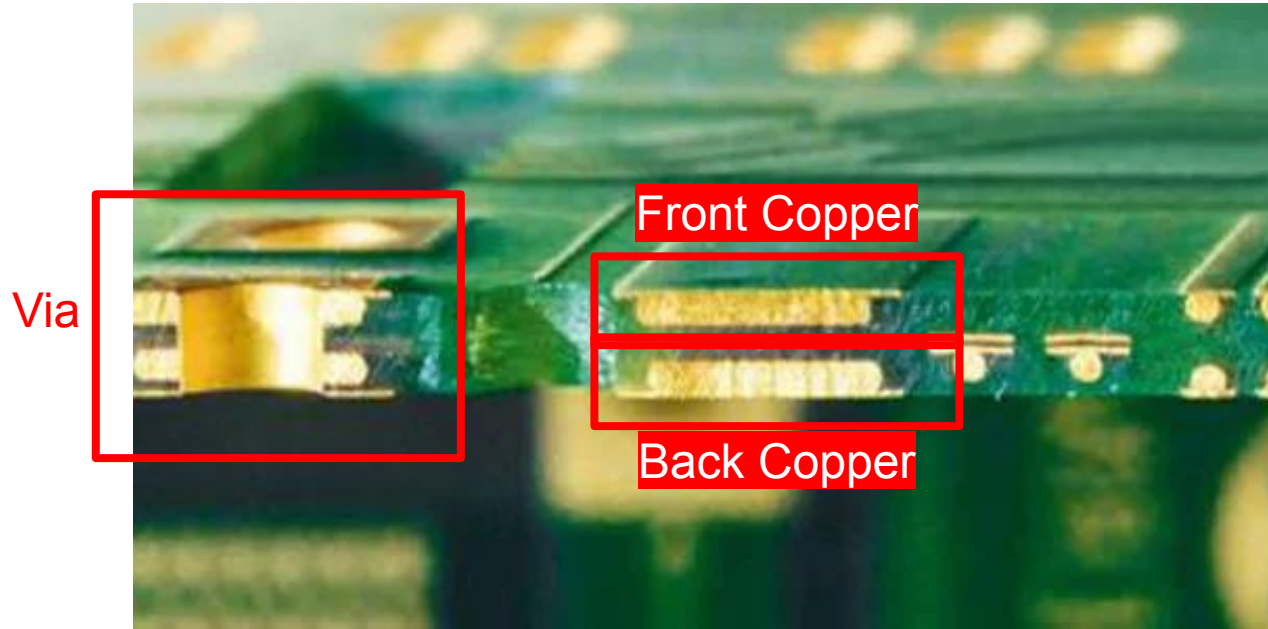
After selecting a component Press (R) to Rotate

Place Traces



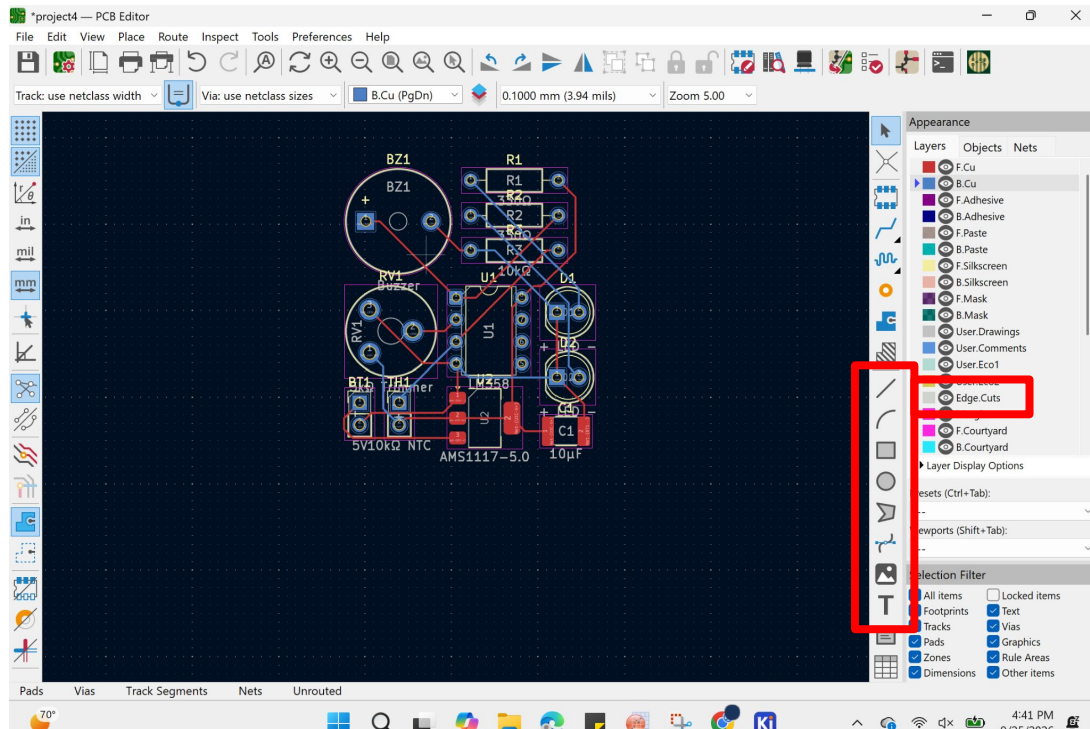
Traces: Shortcut (X) & Vias: (Ctrl+Shift+X)

Place Traces

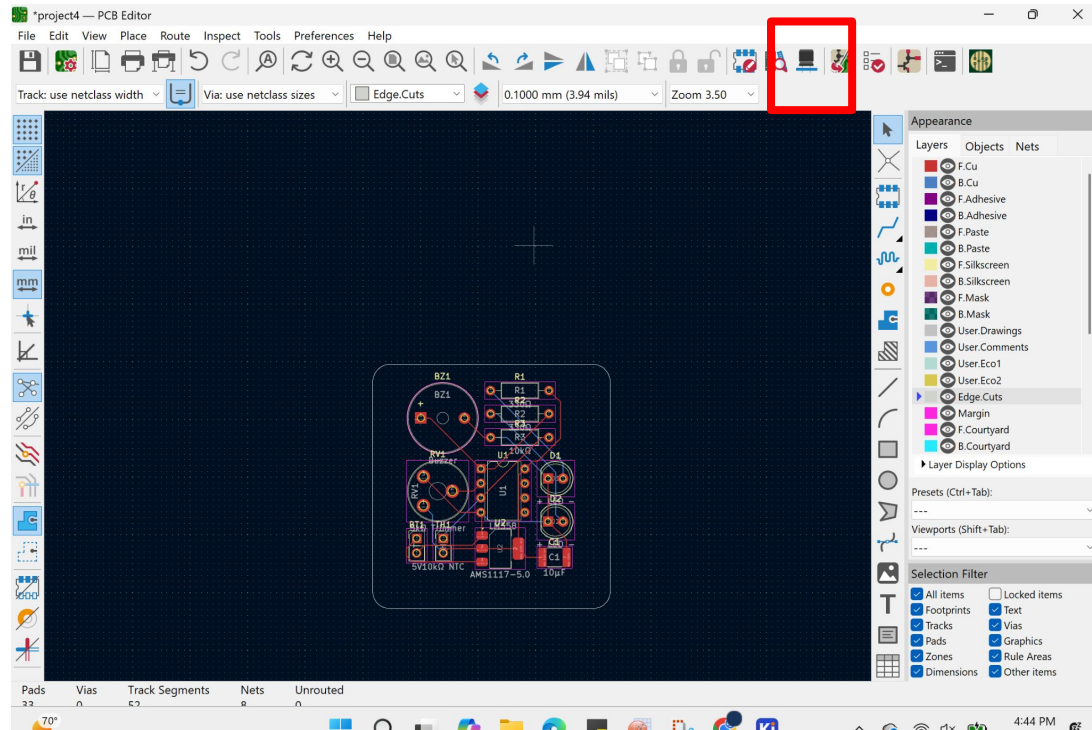


Traces: Shortcut (X) & Vias: (Ctrl+Shift+X)

Placing Cut Lines



3D Viewer



3D Viewer

